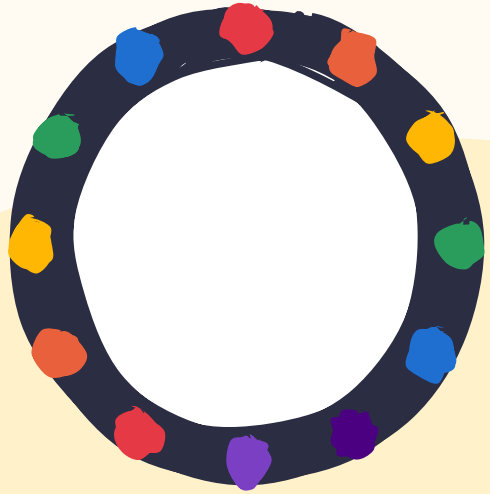
Little Explorers · Class 21

Rainbow LED Ring

Make a circle of lights in every colour!

Rainbow colours **Light colours** **Spin it**



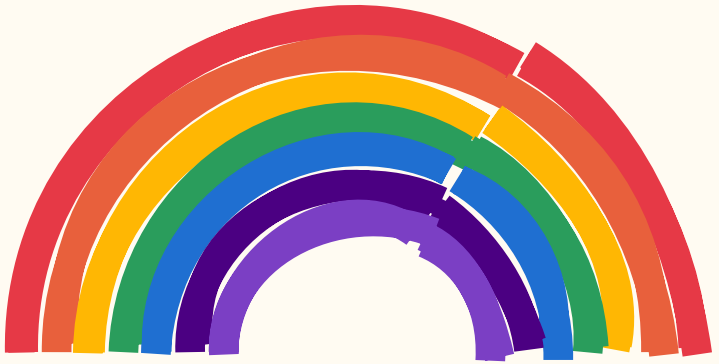
Did you know?
Every colour on a TV is made from tiny red, green and blue lights mixed together.

Let's wonder...

Can we make a rainbow with lights?

No rain needed!

Rainbows appear when the sun shines through rain. But today it's not raining! Could we make our **own** rainbow with tiny lights and a computer?



Paint one?

Wait for rain?

Use colour lights!

Today: you program a **ring of colour lights** to make a rainbow that spins round and round.

Learning objectives

Today I can...

1	**Rainbow colours:** name the **7 rainbow colours** in order	**Proof:** I said the rainbow in order
2	**Light colours:** make the LED ring show **different colours**	**Proof:** my ring showed 3 colours
3	**Spin the rainbow:** make the colours **go round** the ring with a loop	**Proof:** my rainbow spins

Objective 1 · Rainbow colours · video

Video time!



How Is A Rainbow Formed

The Dr. Binocs Show · about 4 minutes · youtu.be/nCPPLhPTAIk

Watch for...

1. What two things make a **rainbow**?
2. How many colours are in a rainbow?
3. What colour is at the top?

Scan to watch

After the video: Say the rainbow colours together: red, orange, yellow, green, blue, indigo, violet!

Objective 1 · Rainbow colours

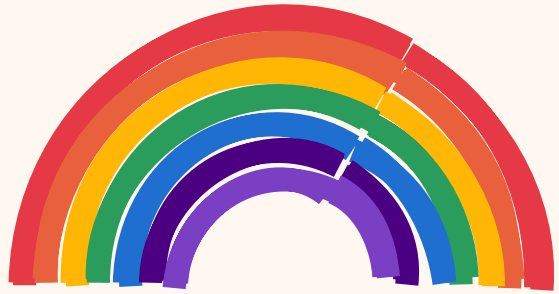
Seven rainbow colours

The idea

Every rainbow has the same colours in the same order: **red, orange, yellow, green, blue, indigo, violet**.

Remember them with a funny name: **ROY G BIV**!

Example: a box of crayons



A crayon box often has the rainbow colours lined up. Our light ring will line them up too, but made of light!

Objective 2 · Light colours

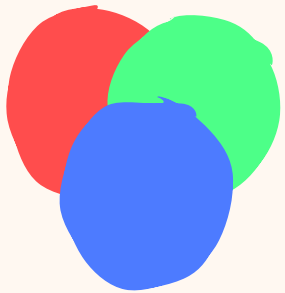
Lights mix colours

The idea

Each light on our ring has **3 tiny lights** inside: red, green and blue. Mix them to make any colour!

- red + green = **yellow**
- red + blue = **purple**
- all three = **white**

Example: torches on a wall

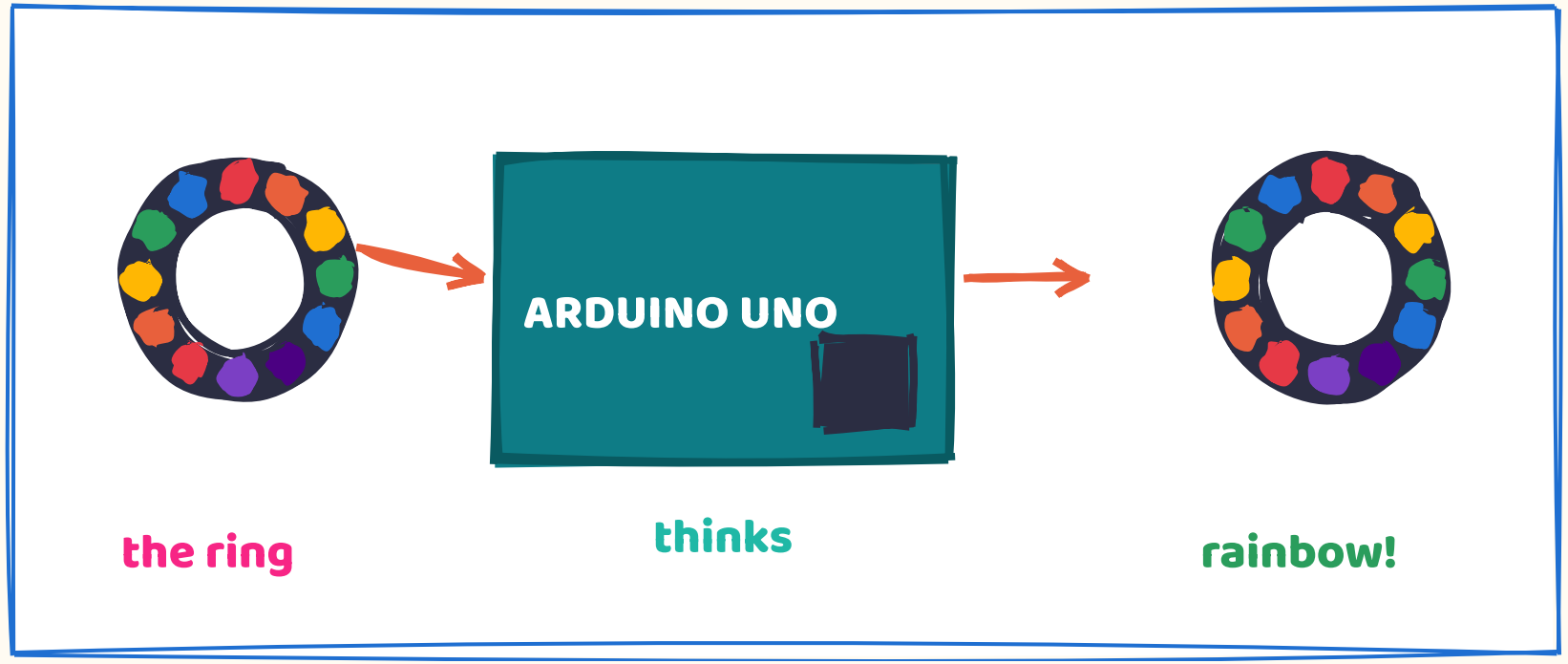


Shine a red torch and a green torch on the same spot of a wall and you see YELLOW! Light mixes differently from paint.

LEARN

Our machine

Think, then show



How it works

The Arduino **thinks**: which light gets which colour?
It sends the colours down a **wire** to the ring.
The lights glow!

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Objective 3 · Spin the rainbow

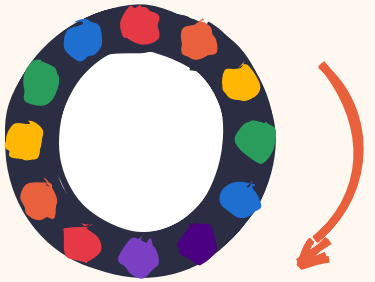
A loop makes it spin

The idea

To make the rainbow **spin**, move every colour one light along, wait a tiny moment, then do it again… **forever**!

The **forever** loop and a short **wait** make it look like the rainbow is running round.

Example: a merry-go-round



On a merry-go-round, every horse moves one place, then another, round and round. Our colours ride the light ring the same way!

Real world

Colour lights everywhere



Festival lights
Eid and wedding lights change colour



Screens
Tiny red, green and blue lights make pictures



Traffic lights
Red, amber and green keep us safe



Concerts
Stage lights spin and flash

? Where have you seen lights that change colour?

Fun fact
A phone screen has millions of tiny colour lights!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human rainbow
2. Snap the picture blocks
3. Snap the picture blocks
4. Speed test
5. Light challenges

Class game

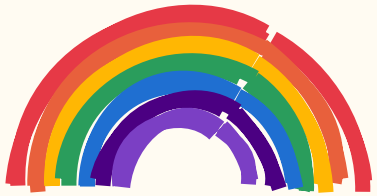
Human rainbow

How to play

1. Seven children each hold a coloured card.
2. Stand in a curve in the right rainbow order.
3. Now everyone moves one place round: the colours 'spin'!
4. Can the class spin the rainbow 3 times?

Then discuss

When the colours moved, did the order change?



Big idea

Moving every colour one place makes a spinning rainbow.

MAKE

Time to build!

Snap the picture blocks



Read it like a story

Each light has a **number**, starting at 0.

Give each number a colour.

Then say **show!**



Teacher

Use ArduinoBlocks with the NeoPixel blocks. Keep brightness low (about 40) so it isn't too bright for eyes.

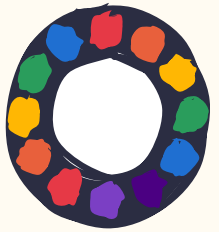
Time to build!

Snap the picture blocks

```
forever
    rainbow starting at colour number
    change start by 1
    show!
    wait 0.1 seconds
```

Read it like a story

Draw the rainbow.
Move it one step.
Wait a little.
Again and again!



Faster?
Change 0.1 to 0.05 to spin faster, or 0.5 to spin slowly.

Experiment!

Speed test

Wait	I think it will be…	It was…
wait 0.5 seconds	slow / fast?	_____
wait 0.1 seconds	slow / fast?	_____
wait 0.02 seconds	slow / fast?	_____

Talk about it

Which speed looked most like magic?

Be kind to eyes

Never stare at bright lights. Keep the ring at arm's length.

MAKE

Level up!

Light challenges

Choose your challenge



Your colour: make the whole ring your favourite colour.



Backwards: make the rainbow spin the other way.



Twinkle: make the lights blink on and off.



Traffic light: red, then amber, then green.

Think like an engineer

How do 3 tiny lights make so many colours?

Boss level

Make the ring look like a clock with a moving hand!

Objective check

Can I really do it? Prove it.

01	Say the 7 rainbow colours in order.	○ ○ ○
02	Show your ring with 3 different colours.	○ ○ ○
03	Show your spinning rainbow. How did you make it faster?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Lights-off show: turn off the classroom lights and show every ring!

Show what you know

What did we learn?

- [x] Rainbows go red, orange, yellow, green, blue, indigo, violet
- [x] Each light mixes red, green and blue
- [x] Each light has a number from 0
- [x] A forever loop makes the rainbow spin

At home
Look for a rainbow in a soap bubble or a CD at home!



Exit ticket: what colour do red and green light make?

Next class:
Button Melody Maker!